

AI-POWERED EMERGENCY AND CRIME REPORTING SYSTEM

INTRODUCTION

What is the Crime Reporting System?

A web-based crime reporting platform enabling citizens to report crimes, seek help, track complaints, and access legal guidance. It also generates heatmap to show crime hotspots in a region and also offers a legal awareness chatbot.

Why is it needed?

It addresses underreporting, delays in emergency response, and the lack of a centralized complaint tracking system.

PROBLEM STATEMENT

- **Delayed Emergency Responses** – Escalates violent crimes and endangers victims.
- **Lack of Complaint Tracking & Transparency** – Discourages reporting.
- **No Legal Awareness Features** – Limits understanding of rights, leading to underreporting.
- **Absence of Heatmap:** Ineffective crime pattern analysis, poor resource allocation, delayed preventive actions, and a lack of public awareness about high-risk areas.

OBJECTIVES

- Web-Based Access
- SOS Emergency Alerts
- Complaint Tracking
- AI-Powered Chatbot
- Heatmap-Based Crime Analysis

EXISTING SYSTEM

Challenges in Existing Online Crime Reporting Systems:

1. Limited Transparency & Complaint Tracking.
2. No AI-Powered Legal Assistance.
3. No Crime Analysis or Hotspot Identification.
4. Limited Integration with Law Enforcement for Quick Response.



WORKING

- **User Registration & Authentication:**

1. Users (citizens and police officers) register using email, phone number, and government ID verification.
2. Secure login with multi-factor authentication ensures data security.

- **Crime Reporting Module:**

1. Users can file complaints by entering crime details, location, and uploading evidence (images/videos).
2. A unique complaint ID is generated for tracking.
3. The complaint is forwarded to the nearest police station based on the user's location.

WORKING

- **Complaint Tracking:**

1. Users can track the status of their complaint through the dashboard.
2. Police officers update the case progress, ensuring transparency.
3. Notifications are sent via email/SMS for every status update.

- **SOS Emergency Assistance:**

1. A dedicated SOS button allows users to send their live location to the nearest police station in critical situations.
2. Integrated Google Maps API & GPS tracking enables police to respond immediately.

WORKING

- **AI-Powered Chatbot for Legal Guidance:**

The chatbot helps users determine:

- Whether an incident qualifies as a crime

- Relevant legal sections and possible actions

- Guidance on how to report crimes properly

Uses LLM-based natural language processing (NLP) to provide accurate responses.

- **Crime Prediction & Hotspot Analysis:**

1. Uses machine learning algorithms to analyze historical crime data.

2. Identifies high-crime areas.

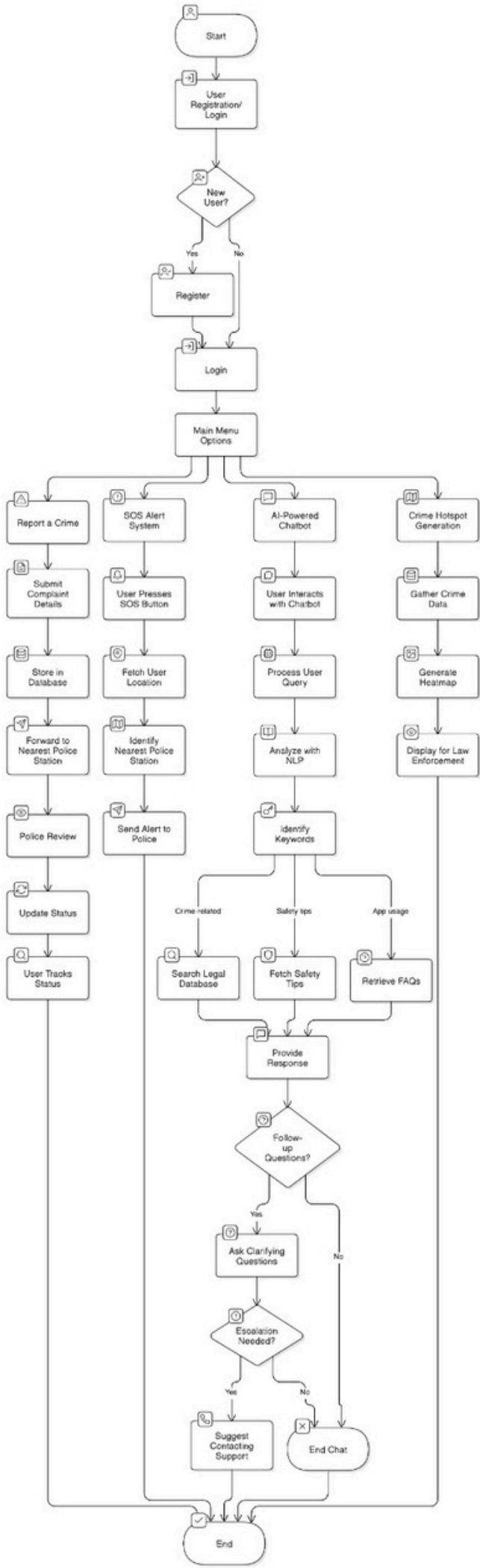
3. Displays a crime heatmap visualization for better insights.

CHATBOT

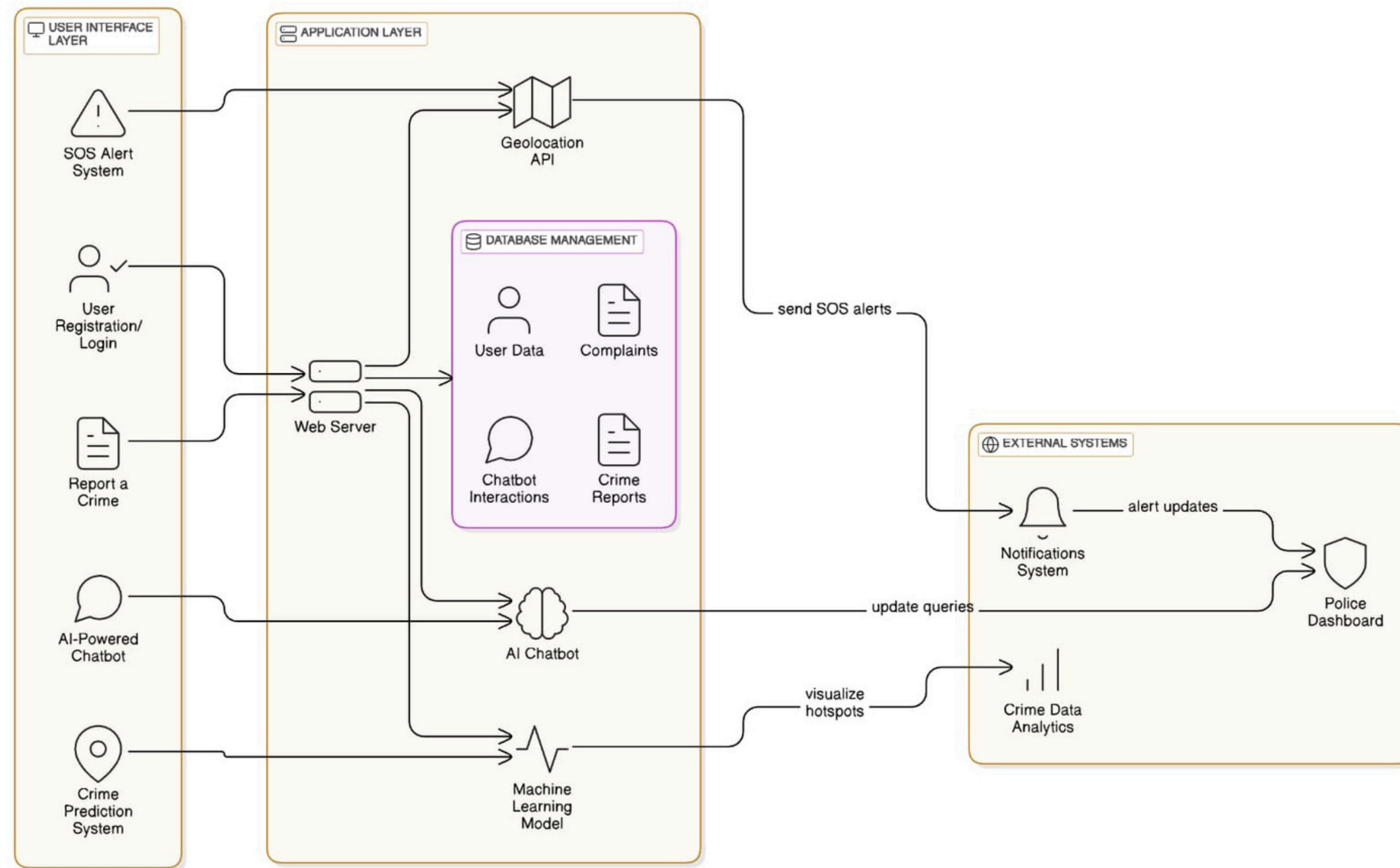
Functionalities:

- Provides information on Indian Penal Code (IPC) sections relevant to different crimes.
- Educates users on the legal procedures for filing complaints and reporting crimes.
- Offers access to past judgments related to specific crimes for legal awareness and case reference.
- Educates users about their fundamental rights as per the Constitution of India to ensure legal empowerment.
- Provides contact details of public prosecutors and lawyers for legal consultation and representation.

FLOW DIAGRAM



ARCHITECTURE DIAGRAM



METHODOLOGY

Front End

HTML, CSS, JavaScript,
React.js

Database

MySQL

Location Tracking (SOS)

Google Maps API

Back End

Python Flask

AI Chatbot

Gemini API

Heatmaps

Folium

IMPLEMENTATION

- **User Registration & Authentication:** JWT and Firebase login for security.
- **Crime Reporting Module:** React.js form for details, file upload, and backend storage with Flask and MySQL.
- **Complaint Tracking System:** Unique complaint ID for tracking; police updates visible on user dashboard.
- **SOS Emergency System:** Google Maps integration for real-time location tracking and GPS coordinates sent to nearest police.
- **AI Chatbot for Legal Assistance:** Gemini AI for NLP-based legal guidance.
- **Crime Analysis and Heatmap Generation:** Visualizing high-risk areas with Folium, enabling authorities to identify patterns and enhance public safety.
- **Admin & Law Enforcement Dashboard:** Police access for complaint review, status updates, and crime heatmap analysis.

CONCLUSION

- **The Crime Reporting System provides a digital, efficient, and AI-powered approach to crime management.**
- **The platform improves public safety, emergency response, and legal awareness.**
- **Machine learning helps in crime prevention by identifying high-risk locations.**

REFERENCES

- **Kaaval Uthavi App link :**

<https://play.google.com/store/apps/details?id=com.amtexsystems.kaavaluthavi>

- **Architecture Diagram Tool :**

Eraser Io

- **Research Paper:**

A Usable and Secure Crime Reporting System

<https://ieeexplore.ieee.org/abstract/document/7096212>

- **Dataset:**

https://eservices.tnpolice.gov.in/content/crime_review/tn_cr_statistics_2022.pdf

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THANK YOU